

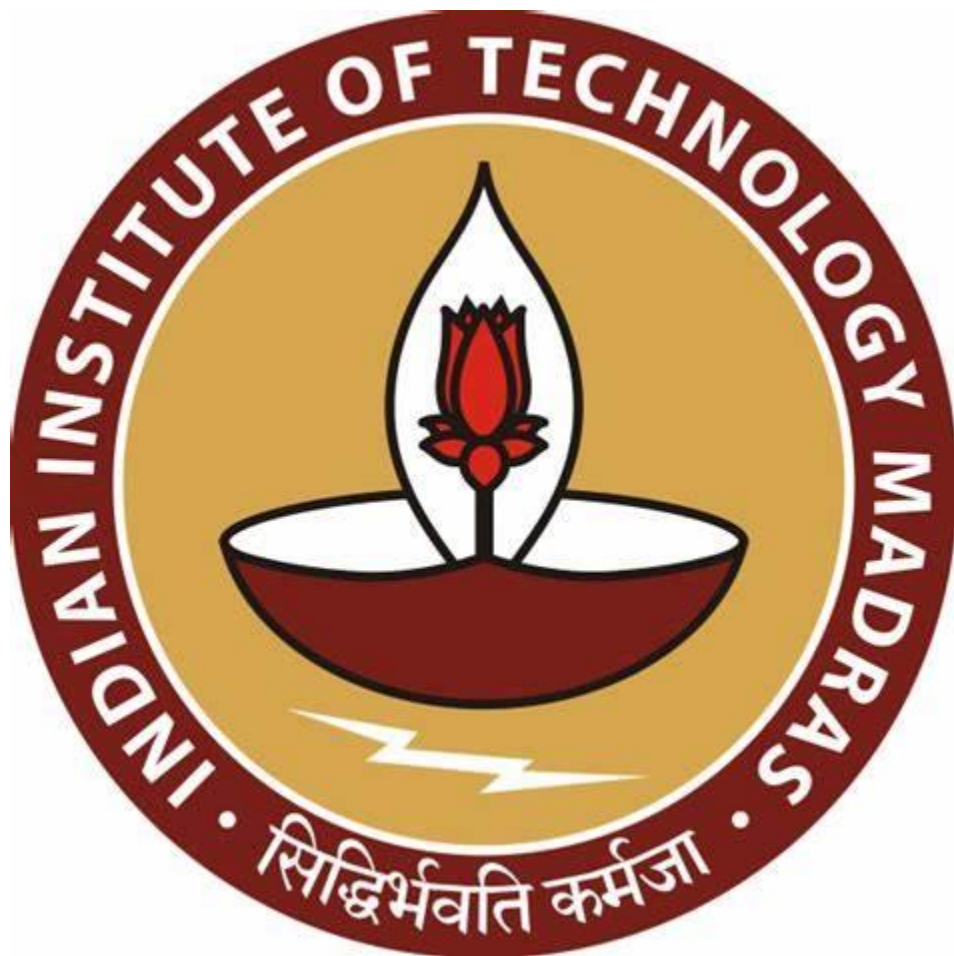
Data-Driven analysis and improvement strategies for swiggy's customer satisfaction

Final Report for the BDM Capstone Project

Submitted By

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Roll No.-22f1000784



Contents

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Declaration Statement

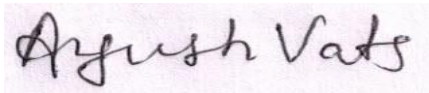
I declare that the project report titled “Data-Driven analysis and Improvement strategies for Swiggy’s customer satisfaction” is my original work conducted as part of my academic requirements at IIT, Madras. Data utilized in this project has been sourced from the Kaggle.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcome and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures. I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of data contained in this project report.

I understand that execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. If plagiarism is detected in the report at any stage of the project evaluation, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project are within the context of the academic project taken up towards course fulfilment in the BS Degree Program offered by the IIT, Madras. The institution doesn’t endorse any of the claims or comments.

Signature of Candidate (Digital Signature)

A handwritten signature in black ink on a light pink background. The signature is written in a cursive style and reads "Ayush Vats".

Name – Ayush Vats

Date – 18-11-2024

Executive Summary

Swiggy, founded in 2014, is India's one of the leading food delivery platform connecting millions of customers with a diverse range of restaurants nationwide. Swiggy primarily operates as B2C(business to consumer) platform.

This report presents an analysis of customer satisfaction issues faced by Swiggy, utilizing a dataset derived from Kaggle. The dataset comprises attributes such as Id, Area, City, Restaurant, Price, Avg ratings, Total ratings, Food type, address and delivery time.

The analysis revealed three key problems faced by Swiggy: low ratings in specific areas, the impact of delivery time on ratings and price sensitivity. After further analysis I found that in low performing areas the average price and delivery time was slightly higher than the high performing areas which led to conclude that delivery time and better prices are significant for customer's satisfaction and suggested ways to do efficient deliveries and price adjusting strategies. Also found that the same food which is performing well in high performing areas are performing bad in low performing area which suggests that the food quality in those areas needs to be improved for satisfying customer.

After further analyzing the impact of delivery time on ratings found that delivery time is crucial for customer's satisfaction. Finally, after analyzing the price sensitivity for customer's satisfaction found that although the increase in prices leads to slightly decrease in the ratings, customers are ready to pay more if the food quality is good.

The recommendations for customer's satisfaction in this report includes strategies like implementing dynamic pricing models, improving delivery efficiency and utilizing customer's feedback for targeted promotions. These strategies aim to enhance service quality and address the identified issues, helping Swiggy improve customer satisfaction and maintain competitive edge in market.

Proof of Originality-Details of repository from where the data was collected

The dataset used in this analysis was sourced from Kaggle, a platform for data science and machine learning. The owner of this dataset created it via web-scrapping and uploaded it on Kaggle. This dataset comprises restaurants-related information from Swiggy. This dataset offers insights into performance metrics of diverse restaurants, essential for comprehending customer preferences and operational challenges in the food delivery sector.

Data Source link - <https://www.kaggle.com/datasets/abhijitdahatonde/swiggy-restuarant-dataset>

Meta Data

The dataset has been downloaded from Kaggle a well-known platform for data-science and machine learning enthusiasts.

General Information about the dataset:

Dataset Name – Swiggy Restaurants Dataset

Total Records(rows) - 7126

Total features(columns) - 10

Purpose – Analyze restaurant delivery data, pricing and ratings to identify patterns for decision-making

Column Details:

Column Name	Description	Non-null count	Data-type	Details
ID	Unique Identifier for each record	7,126	Integer	Acts as the primary key to identify each observation
Area	Area where restaurant is located	7,126	String	Represents locality, used to analyze low ratings in specific areas

City	City in which restaurant is located	7,126	String	High cardinality feature, captures regional patterns
Restaurant	Name of the restaurant	7,126	String	Tracks performance variation among restaurants
Price	Average price for the services provided in the restaurant	7,126	float	Used for evaluating price sensitivity in customer's satisfaction
Avg Ratings	Average customer rating of the restaurant	7,126	float	Primary metric for evaluating performance and identifying problematic areas
Total Ratings	Total number of ratings received by the restaurant	7,126	float	Tracks volume of feedbacks, supports area wise and restaurant wise analyses
Food Type	Type of cuisine or food offered in the restaurant	7,126	String	Highlights preference and performance differences in low vs high performing areas
Address	Full address of the restaurant	7,126	String	Not directly analyzed but supports area – specific filtering
Delivery Time	Estimated delivery time (in minutes)	7,126	float	Key variable in assessing its impact on ratings and customer satisfaction

Descriptive Statistics

The dataset contains 7,126 entries with four numerical columns (excluding ID Column) analyzed in terms of central tendencies and variability measures. The summary is as follows:

Metric	Price	Avg ratings	Total ratings	Delivery time
Count	7,126	7,126	7,126	7,126
Mean	297.80	3.57	70.16	54.06
Standard Deviation	127.65	0.65	28.03	14.13
Minimum	0.0	2.0	20.00	22.00
25 th percentile	200.00	2.90	50.00	44.00
Median (50 th percentile)	300.00	3.70	80.00	52.00
75 th percentile	350.00	4.20	100.00	64.00
Maximum	700.00	5.00	100.00	94

The following table summarizes the key statistics for the categorical columns in the dataset:

Column	Total Entries	Unique Values(count)	Most Frequent value	Frequency of top value
Area	7,126	769	Rohini	243
City	7,126	9	Mumbai	1,076
Restaurant	7,126	6,555	Baskin Robins	10
Food type	7,126	2,987	Indian	361
Address	7,126	1,805	Rohini	213

Detailed explanation of analysis process/methods

Problem 1: Low ratings in Specific areas

Analysis Method:

1. Visualizing the Problem

- A pie chart was plotted to visualize percentage of areas affected by low ratings.
- This helped to identify the proportions of areas with mean avg rating below threshold.
- The threshold for dividing the areas were average rating of 3.5.

2. Comparing Delivery times

- To investigate whether delivery time might influence the rating, mean delivery time was calculated for both areas:

1. Areas with low rating (mean average rating lower than 3.5)

2. Areas with satisfactory rating (mean avg rating greater than 3.5)

Mean delivery time for low performing areas was 56.10 minute and the mean delivery time for high performing area was 52.17 minutes

3. Comparing the mean price

- To investigate whether price is influencing rating in low rating areas, mean price was calculated for both areas:

1. Areas with low rating (mean average rating lower than 3.5)

2. Areas with satisfactory rating (mean avg rating greater than 3.5)

Mean price for low performing areas was 289.0 and for high performing areas was 304.91

4.To see if the difference in the mean delivery time of low performing areas and high performing areas and the difference in average price in low performing areas and high performing area is statistically significant hypothesis testing was Performed.

Method: A two-sample t-test was used because we are comparing the means of two independent groups.

Formula and Calculations:

The test statistic for the t-test is calculated using the formula:

$$t = \frac{(\bar{X}_1 - \bar{X}_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

Where, $\bar{X}_1, \bar{X}_2$: Mean delivery time or mean price of the two group

$(S1)^2, (S2)^2$: is variance of delivery time or variance of prices in two groups

$n1, n2$: is total number of observations in each group

Steps :

1. Compute the sample means of the two group
2. Calculate sample variances
3. Use the above formula to calculate the t-statistics
4. Determine the degrees of freedom (df) using:

$$df = \frac{\left[\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right]^2}{\frac{(s_1^2/n_1)^2}{(n_1 - 1)} + \frac{(s_2^2/n_2)^2}{(n_2 - 1)}}$$

1. Calculated the p-value two-tailed test at significance level ($\alpha = 0.05$) using python library `scipy.stats` to see if the null hypothesis is true or not.

Hypothesis Setup:

Null Hypothesis(H_0): There is no significance difference in mean delivery time or mean price in two groups.

Alternative Hypothesis (H_a): There is significant difference in mean time or mean price in two groups

Interpretation of Results:

If the p-value is less than 0.05, reject null hypothesis and conclude that there is a significant difference in mean delivery times or mean prices between the two groups.

If the p-value is greater than 0.05, fail to reject H_0 , indicating that the difference in mean delivery time or mean prices is not statistically significant

5. Comparing the performance of Food types across both regions (low performing areas and high performing areas)

- First Calculated the 10 most frequent food types in both the reason
- Then, Plotted a bar graph showing average rating for food types in both the region to see the performance of the same food type in both the region

Problem 2: Impact of delivery time on ratings

1. Plotted a scattered plot (delivery time vs. Avg ratings) to see if there is any relationship between the both variables.
2. Calculated the Pearson correlation coefficient to assess the strength and direction of linear relationship between the delivery time and avg ratings. It is calculated using the below formula:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

Where:

x and y are the two variables compared

n is the number of data points

3. I divided the ratings into three categories low, medium, high. Avg ratings between 0-2.5 fell into low category, 2.5-3.5 fell into medium category and above 3.5 fell into high category
4. After dividing the categories I plotted Box plot (Delivery time distribution across different categories)

Problem 3 Price sensitivity in customer's satisfaction

1. Plotted a scatter plot (price vs average ratings) to visualize the linear relationship between the price and average ratings. It is calculated using the below formula:
2. Calculated the Pearson correlation coefficient to assess the strength and direction of linear relationship between price and average ratings

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

Where x,y are the two variables compared

And n is the number of data points

3. To analyze further I plotted a bar chart (price range vs average ratings) by dividing the prices into 4 ranges 0-100, 100-200, 200-300 and 300+ which helped me to visualize the average ratings in each price range.

Results and findings

Problem 1 low ratings in specific areas

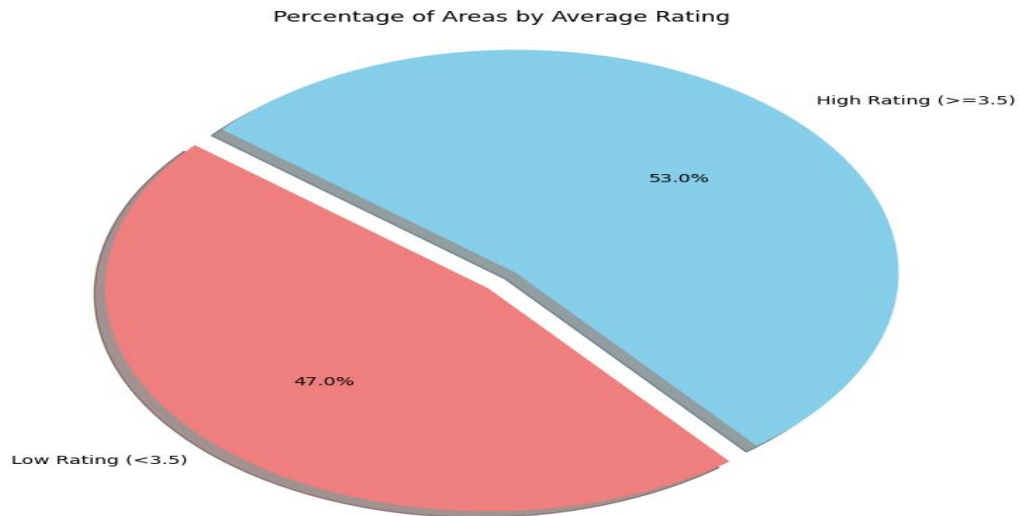


Figure 1.0

1. Above pie chart fig- 1.0 represents the percentage of areas with with mean average ratings higher than 3.5 and lower than 3.5
2. From the pie chart we can clearly see that there is huge percentage of area (47.07%) that falls within the low performing area (areas with mean avg ratings below 3.5)
3. After Calculating the mean delivery time for low performing areas (areas with mean average rating ratings lower than 3.5) it came to be **56.10** and the mean delivery time for high performing area (areas with mean average rating higher than 3.5) came to be **52.17**.
4. Since the difference between the mean delivery time in both region is only 3.93. I conducted hypothesis testing to find if the difference is statistically significant or not.

After calculating the t-statistic using formula

$$t = \frac{(\bar{X}_1 - \bar{X}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

The value of t-statistic , t = -11.85

The p-value = 0.00

Since, the p-value is lesser than 0.05 we reject the null hypothesis and conclude that the difference in delivery time between the high performing areas and low performing areas is statistically significant.

4. After calculating mean prices for low performing and high performing area the mean price for low performing areas comes as Rs. **289.77** and for high performing areas comes as **Rs. 304.91**
5. To see if the difference between mean prices in two categories is statistically significant hypothesis testing was performed:

T-statistic, t = -5.00

P-value = 0.00

Since the p-value is below 0.05 the null hypothesis is rejected and we conclude that the difference in mean price between high performing and low performing area is statistically significant.

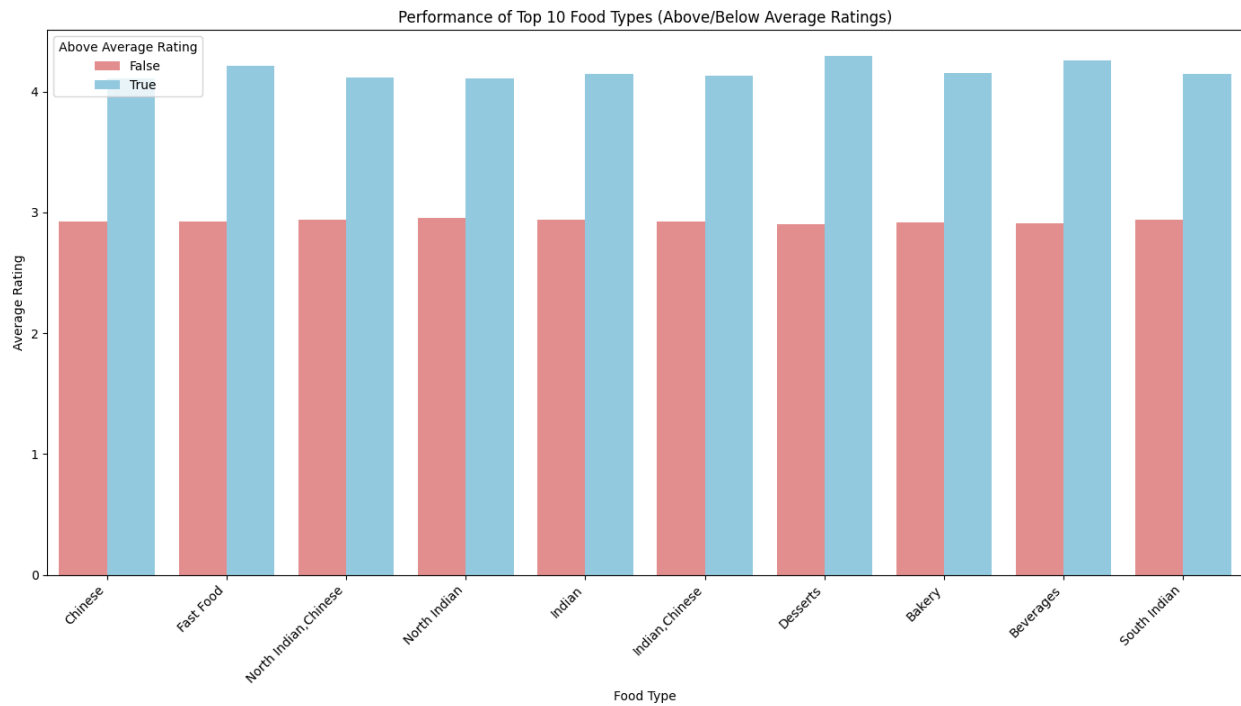


Fig- 2.0

6. To compare the performance of top 10 food types in both high performing and low performing area plotted a bar graph

7. From the figure 2.0 we can easily conclude that the foods in high performing areas are performing way better than low performing area.

Problem 2: Impact of delivery time on ratings

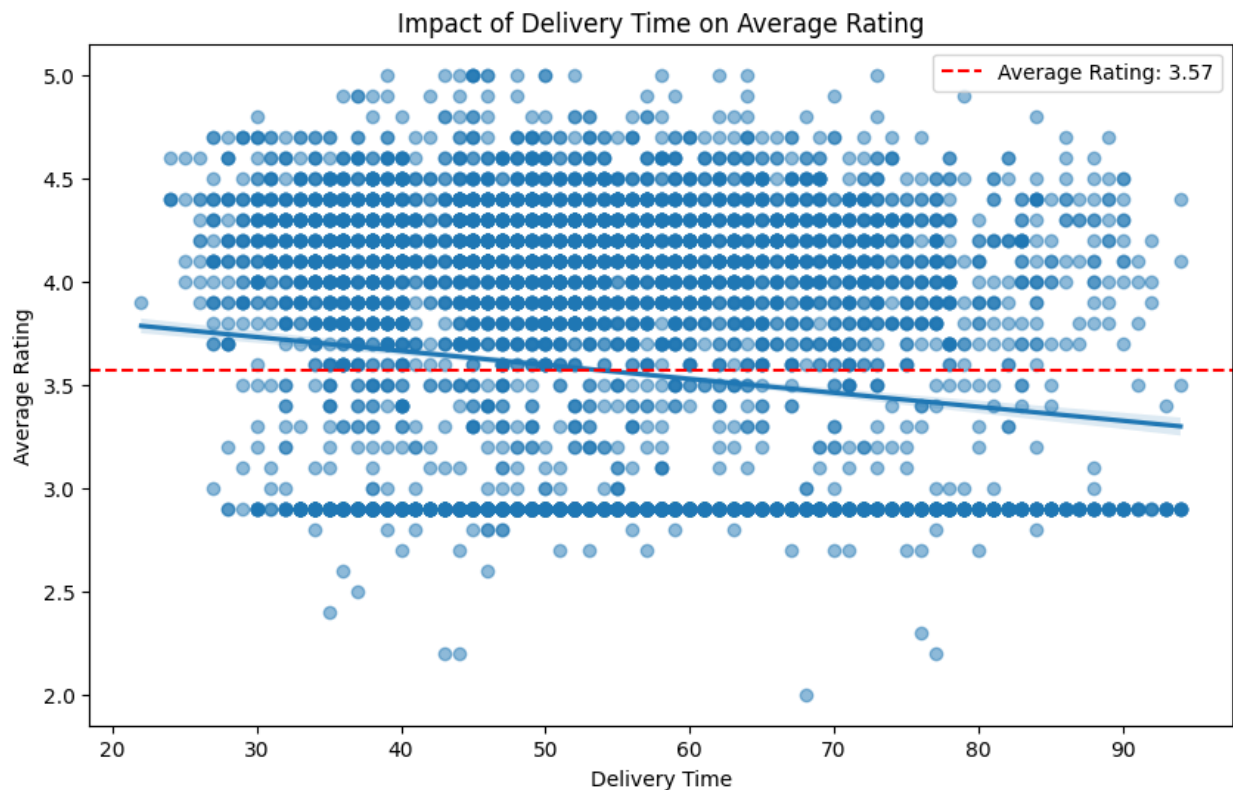


Fig: 3.0

1. Scatter plot(delivery time vs avg ratings) was plotted to visualize relationship between delivery time and average ratings

Overall Trend:

The scatter plot illustrates a negative correlation between delivery time and average rating. This means that as delivery time increases, the average rating tends to decrease.

Key Observations:

Negative Correlation: The downward sloping trend of the scatter points confirms the negative correlation between delivery time and average rating.

Clustering: The data points are clustered around the trend line, indicating a moderate to strong relationship.

2. Calculate Pearson correlation coefficient to see the strength of linear relationship between delivery time and avg ratings, it came as **0.15**.

- The value 0.15 indicates a weak negative correlation between the two variables
- It implies a small increase in delivery time leads to slight decrease in avg ratings.

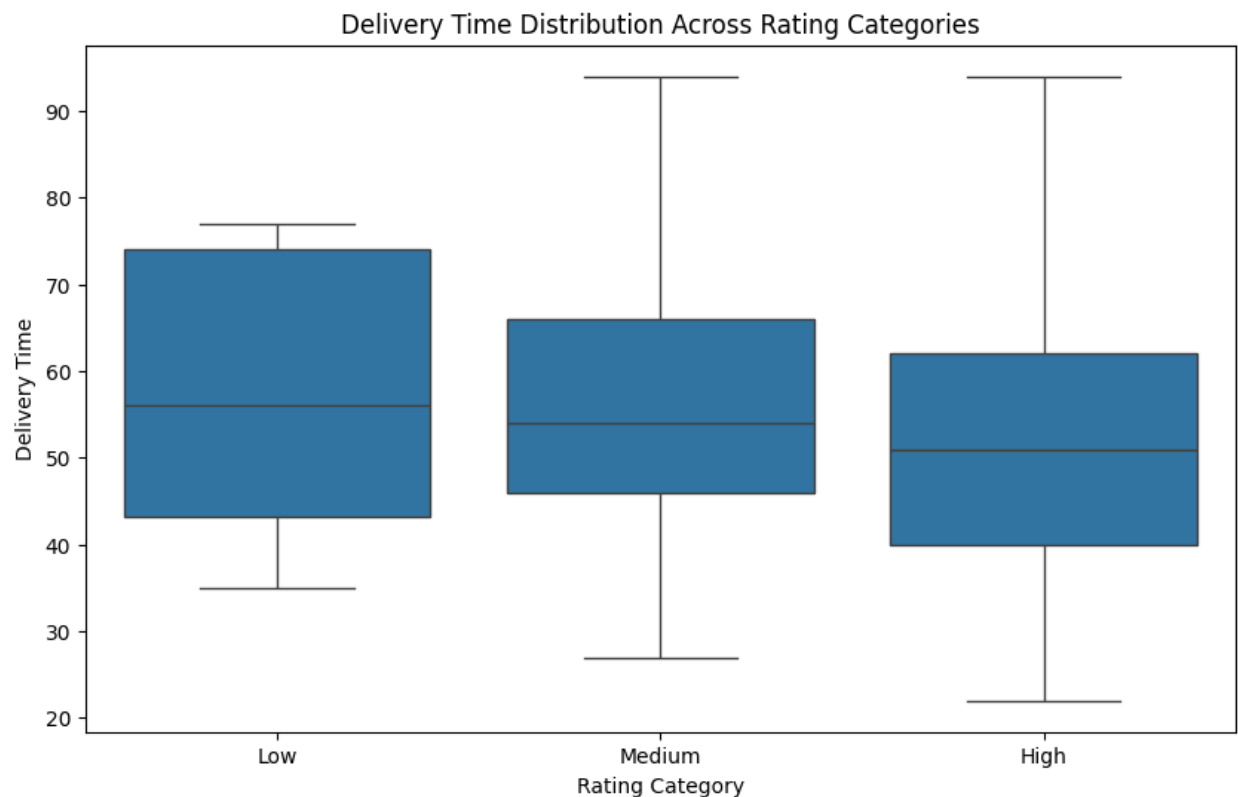


Fig- 4.0

Overall Trend:

The box plot displays the distribution of delivery time across three rating categories: Low (0-2.5), Medium (2.5-3.5), and High (3.5-5.0).

Key Observations:

Median Delivery Time: The median delivery time for the "Low" rating category is the highest, followed by the "Medium" and "High" categories. This suggests that longer delivery times are associated with lower ratings.

Range of Delivery Times: The "Low" category has the widest range of delivery times, indicating greater variability in delivery times for orders with lower ratings. The "High" category has the narrowest range, suggesting more consistent delivery times for orders with higher ratings.

Problem 3: Price sensitivity

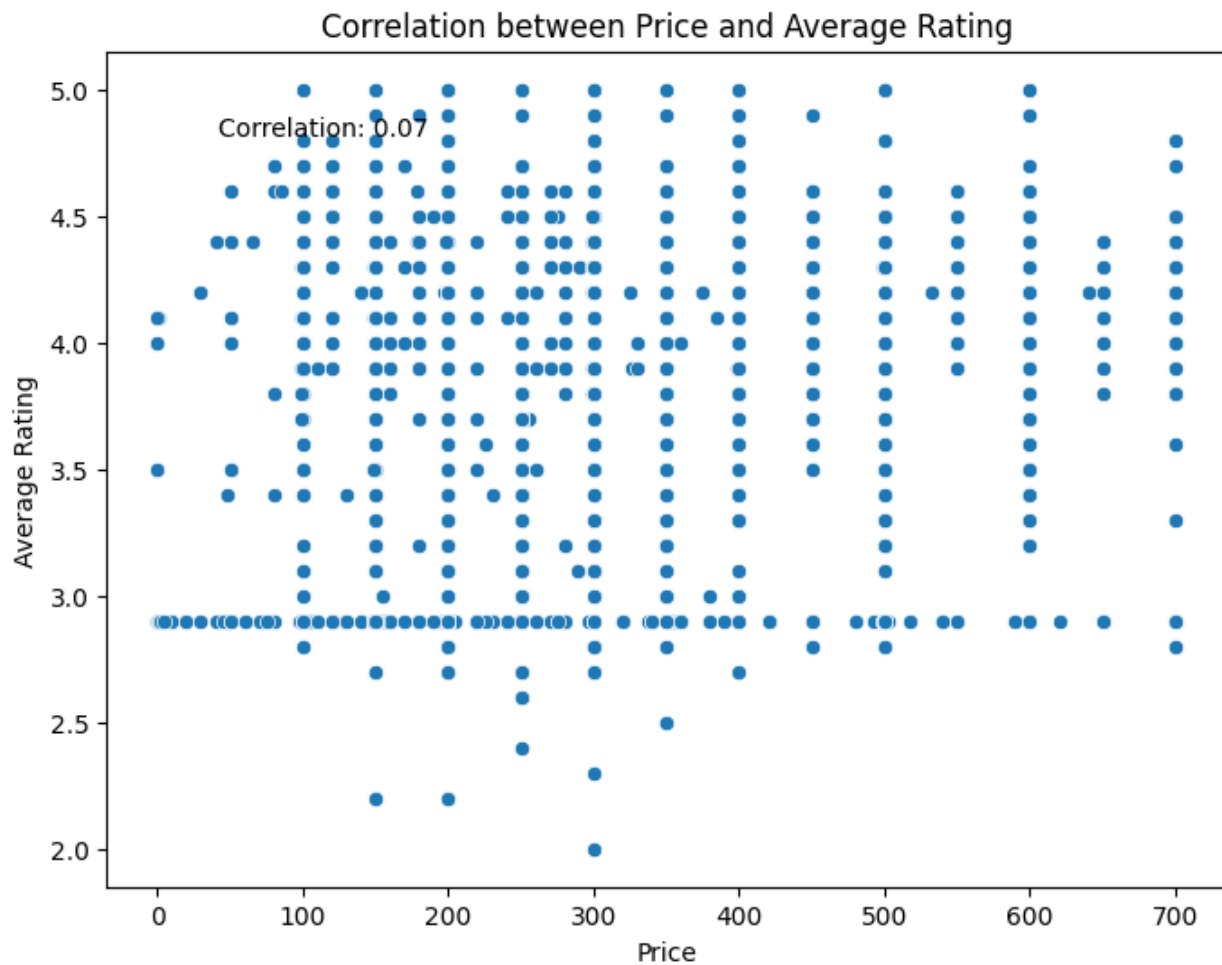


Fig: 5.0

1. Correlation:

The correlation coefficient displayed on the plot is 0.07. This indicates a very weak positive correlation between price and average rating. In other words, there is almost no relationship between the price of a product and its average rating.

Key Observations:

Scattered Data Points: The data points are widely scattered across the plot, with no clear pattern or trend. This further supports the conclusion of a weak correlation.

No Consistent Relationship: There are data points with both high and low prices across all rating categories. This suggests that price alone is not a strong predictor of average rating.

2. Calculated Pearson correlation coefficient to see the strength of linear relationship between price and avg ratings, it came as **0.0680**

This is very close to 0, which suggests that there is almost no linear relationship between the two variables



Fig: 6.0

Overall Trend:

The bar chart shows the average rating for products within four different price ranges: 0-100, 100-200, 200-300, and 300+. We can observe a general trend of increasing average ratings with higher price ranges.

Key Observations:

Increasing Trend: The average rating increases as the price range increases. The highest average rating is observed for the 300+ price range.

Moderate Differences: While the trend is clear, the differences in average ratings between the price ranges are not very large.

Interpretation of results and recommendation

Problem 1: Low ratings in specific areas

The statistical analysis reveals significant differences in both delivery time and price between high-performing and low-performing areas.

- **Delivery Time:** High-performing areas have significantly lower average delivery times compared to low-performing areas. This suggests that delivery efficiency and logistics are key factors influencing customer satisfaction and ratings in low-performing areas.
- **Price:** High-performing areas have slightly higher average prices compared to low-performing areas. While this might be due to factors like higher-quality food or premium offerings, it's important to ensure that the price difference is justified by the overall experience.

Food Performance Comparison:

The bar graph comparison of top 10 food types highlights the performance disparity between high-performing and low-performing areas. This indicates that specific food items or categories might be contributing to the lower ratings in certain areas.

Recommendations for Swiggy:

1. Targeted Delivery Optimization:

- a. **Real-time Tracking and Alerts:** Implement a robust system to track orders in real-time and provide timely updates to customers.
- b. **Optimized Delivery Routes:** Utilize advanced algorithms to optimize delivery routes and reduce delivery times.
- c. **Dedicated Delivery Partners:** Consider partnering with dedicated delivery partners who prioritize timely deliveries and customer satisfaction.

2. Quality Control and Standardization:

- a. **Strict Quality Standards:** Enforce stringent quality standards for all food items, ensuring consistency across all outlets.
- b. **Regular Audits and Inspections:** Conduct regular audits and inspections to maintain high quality standards.

- c. **Training and Development:** Provide comprehensive training to restaurant partners on food preparation, hygiene, and packaging.
- 3. **Customer Experience Enhancement:**
 - a. **Personalized Recommendations:** Utilize data analytics to offer personalized recommendations based on customer preferences.
 - b. **Efficient Customer Support:** Provide efficient and responsive customer support channels to address concerns and complaints promptly.
 - c. **Loyalty Programs and Rewards:** Implement loyalty programs to incentivize repeat customers and reward customer loyalty.
- 4. **Data-Driven Decision Making:**
 - a. **Regular Data Analysis:** Conduct regular analysis of customer feedback, ratings, and delivery metrics to identify areas for improvement.
 - b. **Predictive Analytics:** Utilize predictive analytics to forecast demand, optimize inventory, and improve delivery planning.
- 5. **Targeted Marketing and Promotions:**
 - a. **Localized Marketing Campaigns:** Tailor marketing campaigns to specific areas, highlighting unique offers and promotions.
 - b. **Social Media Engagement:** Actively engage with customers on social media to address concerns and build brand loyalty.

By implementing these recommendations, Swiggy can significantly improve customer satisfaction and ratings in low-performing areas, leading to overall business growth and success.

Problem 2: Impact of delivery time on ratings

Pearson Correlation Coefficient of 0.15:

While there is a negative correlation, the weak correlation coefficient suggests that delivery time is not the sole determinant of average ratings. Other factors like food quality, packaging, and customer service also play significant roles.

Recommendations:

1. **Prioritize Faster Deliveries:** Continue to optimize delivery routes, increase delivery fleet size, and explore partnerships with faster delivery services to reduce delivery times.
2. **Real-time Tracking and Updates:** Implement a robust system to track orders in real-time and provide timely updates to customers.
3. **Transparent Communication:** Keep customers informed about any potential delays and provide proactive updates.

Delivery Time Distribution Across Rating Categories

The box plot highlights the relationship between delivery time and rating categories:

- **Lower Ratings, Longer Delivery Times:** The "Low" rating category exhibits the highest median delivery time, indicating a strong correlation between longer delivery times and lower customer satisfaction.
- **Higher Ratings, Shorter Delivery Times:** The "High" rating category shows the lowest median delivery time, suggesting that faster deliveries contribute to higher ratings.

Recommendations:

1. **Targeted Delivery Optimization:** Focus on improving delivery times in low-performing areas to enhance customer satisfaction and ratings.
2. **Incentivize Timely Deliveries:** Implement incentives for delivery partners to prioritize timely deliveries and reduce delays.
3. **Data-Driven Insights:** Utilize data analytics to identify areas with consistently long delivery times and implement targeted solutions.

Problem 3: Price sensitivity

Price vs. Average Rating

Weak Correlation:

The low correlation coefficient (0.07 and 0.0680) indicates that price alone is not a strong predictor of average ratings. This suggests that other factors, such as product quality, customer service, and brand reputation, play a more significant role in determining customer satisfaction.

Recommendations:

1. **Focus on Quality and Service:** Prioritize offering high-quality products and exceptional customer service, regardless of price.
2. **Value Perception:** Clearly communicate the value proposition of products, highlighting unique features, benefits, and brand reputation.
3. **Data-Driven Pricing:** Utilize data analytics to optimize pricing strategies based on customer preferences, competitor pricing, and product cost.

Price Range vs. Average Rating

Positive Trend:

The bar chart reveals a positive trend between price range and average rating. Higher-priced products tend to have higher average ratings.

Potential Explanations:

1. **Premium Perception:** Higher-priced products may be perceived as premium, leading to higher expectations and, consequently, higher ratings.
2. **Product Quality:** Higher-priced products might offer superior quality, features, or ingredients, contributing to better customer satisfaction.
3. **Target Market:** Higher-priced products may be targeted towards a specific customer segment with higher expectations and willingness to pay.

Overall Recommendations:

To improve customer satisfaction and ratings across all price ranges, Swiggy should:

1. **Prioritize Customer Experience:** Focus on providing excellent customer service, timely deliveries, and accurate order fulfillment.
2. **Leverage Data Analytics:** Utilize data to identify trends, optimize operations, and make data-driven decisions.
3. **Continuous Improvement:** Implement a culture of continuous improvement, regularly evaluating and refining processes.
4. **Transparent Communication:** Maintain open and transparent communication with customers, addressing concerns and feedback promptly.

By focusing on these key areas, Swiggy can enhance customer satisfaction, increase ratings, and drive business growth.

Presentation and legibility report

1.1 Structure and Organization

- **Title Page:** The title page includes the project title, student name, roll number, institution, and contact details. This provides a clear identification of the project and its author.
- **Contents Page:** The table of contents is well-organized, listing sections such as the Declaration Statement, Executive Summary, Proof of Originality, Meta Data and Descriptive Statistics, Detailed Explanation of Analysis Process, Results and Findings, Interpretation of Results and Recommendations, and Presentation and Legibility Report. This structure aids in easy navigation through the document.
- **Headings and Subheadings:** Each section is clearly labeled with headings and subheadings, ensuring that readers can easily find specific information. Headings are formatted in bold and slightly larger fonts for emphasis.

- **Figures and Tables:** All figures and tables are appropriately labeled and referenced in the text. Descriptive captions are provided for clarity, and tables are formatted for easy readability with clear delineation of rows and columns.
- **Consistency:** The formatting throughout the report is consistent, with uniform font styles and sizes, consistent use of bullet points, and alignment. This uniformity enhances readability and professionalism.

1.2 Visual Design

- **Font and Size:** The report uses a standard font (e.g., Arial) in a readable size (e.g., 11 pt) for the main text. Headings are in a slightly larger font (e.g., 12 pt) to distinguish sections.
- **Color Scheme:** The report uses a minimal color scheme to maintain a professional appearance. Text is primarily in black, with any highlights or emphasis provided in a subtle color that does not distract from the content.
- **Graphs and Charts:** All graphical elements, including bar charts, pie charts, and scatter plots, are clearly labeled with legends and axes titles. Colors used in graphs are distinct and contrast well with the background for ease of interpretation.

1.3 Clarity and Readability

- **Language:** The language used in the report is clear and concise. Technical terms are explained where necessary, and complex sentences are avoided to ensure that the content is accessible to a broad audience.
- **Summarization:** Key findings and recommendations are summarized at the end of relevant sections. This helps readers quickly grasp the main points without needing to read through extensive text.
- **Tables and Data:** Tables are organized logically with clear headings and units. Data is presented in a structured manner, making it easy to compare and interpret results.

2. Legibility

2.1 Text Formatting

- **Spacing:** The report uses appropriate line spacing (e.g., 1.5 or double spacing) to ensure readability. Paragraphs are well-separated, and there is adequate spacing between sections.
- **Alignment:** Text is left-aligned for consistency, and important sections such as headings and subheadings are centered or bolded to stand out.
- **Margins:** Standard margins (e.g., 1 inch on all sides) are maintained throughout the report, ensuring that text is not cramped and that there is space for annotations if needed.

2.2 Document Quality

- **Proofreading:** The report has been thoroughly proofread to eliminate grammatical errors, typos, and inconsistencies. Attention to detail in proofreading enhances the overall quality of the document.
- **References:** Sources and references are cited appropriately throughout the document. A reference list is provided at the end of the report, formatted according to academic standards (e.g., APA or MLA).

2.3 Accessibility

- **Text Accessibility:** Text is formatted to be easily readable, with appropriate font sizes and contrast. There is no use of overly complex language or jargon without explanation.
- **Data Accessibility:** Data presented in tables and graphs is clearly labeled and explained. Any abbreviations or symbols used are defined in a legend or key.